

Answers for React Tutorial Exercises

Setup and Warm-Up

Step: Ensure the app runs successfully.

Command to start the app:

npm start

- - The default screen should display **React's welcome page**.
-

Exercise 1: Create a Functional Component

Answer: `Welcome.js`

```
import React from 'react';
```

```
function Welcome() {  
  return <h1>Welcome to React Tutorial!</h1>;  
}
```

```
export default Welcome;
```

App.js Update:

```
import React from 'react';  
import Welcome from './Welcome';
```

```
function App() {  
  return (  
    <div>  
      <Welcome />  
    </div>  
  );  
}
```

```
export default App;
```

Exercise 2: Passing Props

Answer: Updated `Welcome.js`

```
import React from 'react';

function Welcome(props) {
  return <h1>Welcome to React Tutorial, {props.name}!</h1>;
}

export default Welcome;
```

App.js Update:

```
import React from 'react';
import Welcome from './Welcome';

function App() {
  return (
    <div>
      <Welcome name="Alice" />
      <Welcome name="Bob" />
    </div>
  );
}

export default App;
```

Exercise 3: Create a List with Components

Answer: `TodoList.js`

```
import React from 'react';

function TodoList() {
  return (
    <div>
      <h2>My Tasks:</h2>
      <ul>
        <li>Learn React</li>
        <li>Build Components</li>
        <li>Pass Props</li>
      </ul>
    </div>
  );
}
```

```

        </ul>
      </div>
    );
  }

export default TodoList;

```

App.js Update:

```

import React from 'react';
import TodoList from './TodoList';

function App() {
  return (
    <div>
      <TodoList />
    </div>
  );
}

export default App;

```

Extension: Use Props for Tasks

- Updated `TodoList.js`:

```

import React from 'react';

function TodoList(props) {
  return (
    <div>
      <h2>My Tasks:</h2>
      <ul>
        {props.tasks.map((task, index) => (
          <li key={index}>{task}</li>
        ))}
      </ul>
    </div>
  );
}

export default TodoList;

```

- Updated `App.js`:

```
import React from 'react';
import TodoList from './TodoList';

function App() {
  const tasks = ["Learn React", "Build Components", "Pass Props"];

  return (
    <div>
      <TodoList tasks={tasks} />
    </div>
  );
}

export default App;
```

Exercise 4: Using JSX Expressions

Answer: Update `TodoList.js` to include task count:

```
import React from 'react';

function TodoList(props) {
  const taskCount = props.tasks.length;

  return (
    <div>
      <h2>You have {taskCount} tasks:</h2>
      <ul>
        {props.tasks.map((task, index) => (
          <li key={index}>{task}</li>
        ))}
      </ul>
    </div>
  );
}

export default TodoList;
```

App.js remains unchanged.

Exercise 5: Styling Components

Answer:

1. Create a `styles.css` file:

```
.todo-list {  
  background-color: #f0f8ff;  
  padding: 20px;  
  border-radius: 5px;  
}
```

```
.task {  
  font-weight: bold;  
  font-style: italic;  
}
```

2. Apply styles in `TodoList.js`:

```
import React from 'react';  
import './styles.css';  
  
function TodoList(props) {  
  const taskCount = props.tasks.length;  
  
  return (  
    <div className="todo-list">  
      <h2>You have {taskCount} tasks:</h2>  
      <ul>  
        {props.tasks.map((task, index) => (  
          <li key={index} className="task">{task}</li>  
        ))}  
      </ul>  
    </div>  
  );  
}  
  
export default TodoList;
```

Wrap-Up

By the end of the session, the app should:

1. Show a greeting with `Welcome.js`.
2. Dynamically display a name via props.
3. Render a task list with a count and apply styling.

Students can further experiment with states and interactivity!